

Bayesian Persuasion

Correct Forecasts, Conflicting Objectives

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- ▶ We will characterize how much control this gives the provider and use a geometric argument to find its best disclosure policy.
- ▶ Perfectly calibrated reports need not reveal the information that would be most useful to the person making the decision.

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The provider is the *sender*: it earns 1 for deployment and 0 for holding. Which information will it choose to reveal?

The User Needs an 80% Safety Probability

Write x for the user's probability of safety given the information she has received—her *posterior* safety probability.

$$\mathbb{E}[u_R(\text{deploy}, \theta) \mid \text{information}] = x \cdot 1 + (1 - x)(-4) = 5x - 4.$$

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We assume the user deploys at equality, when the two actions tie. Write q for a generic safety threshold; for this user, $q = 4/5$.

Can Partial Disclosure Induce More Deployment?

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Can the provider induce more deployment than full disclosure does, even when the user knows how its reports are produced?

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For AI providers, costly training can make model updates slower than users' adaptation to deployed behavior. This motivates treating a fixed model and reporting protocol as a commitment during that period.

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Send d (recommend deployment) for every safe proposal. For a harmful proposal, send d with probability r and h (recommend holding) otherwise.

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Solving for r gives

$$\frac{2}{5} + \frac{3}{5}r = \frac{2/5}{4/5} = \frac{1}{2}, \quad r = \frac{1/2 - 2/5}{3/5} = \frac{1}{6}.$$

This constructs a feasible rule; we have not yet shown it is optimal.

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After d , the user assigns safety probability $4/5$ and deploys. Signal h occurs only when harmful, so she holds. Deployment rises from $2/5$ under full disclosure to $1/2$.

The rule *pools* all safe proposals and some harmful ones into the same signal. It never claims that deployment is certainly safe.

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Disclosure policy	Deployment probability	User utility
No information	0	0
Full disclosure	$2/5$	$2/5$
Provider's rule	$1/2$	0

The user loses relative to *full disclosure*, not relative to receiving no information. She could always ignore a signal.

The Reports Are Nevertheless Calibrated

Suppose the provider reports the implied safety probability:
 $\hat{p} = 4/5$ after d , and $\hat{p} = 0$ after h . Our calculation gives

$$\mathbb{P}(\text{safe} \mid \hat{p} = 4/5) = 4/5, \quad \mathbb{P}(\text{safe} \mid \hat{p} = 0) = 0.$$

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Lecture 9's lesson applies: the guarantee depends on which information we condition on. These reports are calibrated given the reported probability, but not given everything the provider knows.

A Finite Model of Information Design

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Let Θ be a finite state set and $\mathcal{A}$ a finite set of receiver actions.

- ▶ $b_0 \in \Delta(\Theta)$ is the common prior; $\Delta(\Theta)$ denotes the probability distributions on Θ .
- ▶ Discard states of prior probability zero. Thus $b_0(\theta) > 0$ for every remaining state (*full support*).
- ▶ $u_R(a, \theta)$ and $u_S(a, \theta)$ are receiver and sender utilities. The prior and utilities are known to both parties.

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The sender chooses a finite set of signals $\mathcal{S}$ and a rule

$$\sigma(s \mid \theta) \geq 0, \quad \sum_{s \in \mathcal{S}} \sigma(s \mid \theta) = 1 \quad \text{for each } \theta.$$

The sender observes the realized state. The receiver observes only the report generated according to the committed signaling rule.

A Signal Has a Probability and a Posterior

Fix a signaling scheme. The probability of signal s is

$$\alpha_s := \mathbb{P}(s) = \sum_{\theta \in \Theta} b_0(\theta) \sigma(s \mid \theta).$$

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An experiment therefore produces a collection of belief vectors b_s , each occurring with probability α_s . This notation will let us average entire belief distributions.

Which such collections can the sender create?

The Mean Posterior Is the Only Restriction

A *posterior split* specifies finitely many beliefs b_s and positive weights α_s summing to one. It is *Bayes plausible* if

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Theorem (Splitting lemma)

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Two claims:

1. Bayesian posteriors must average to the prior.
2. Every split with that mean can be implemented by a public rule.

Necessity: Beliefs Average Back to the Prior

Fix a state θ . For each signal with $\alpha_s > 0$, Bayes' rule gives

$$b_s(\theta) = \mathbb{P}(\theta \mid s) = \frac{b_0(\theta)\sigma(s \mid \theta)}{\alpha_s}.$$

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The equality holds in every state coordinate θ , so the average posterior vector equals the prior vector: $\sum_s \alpha_s b_s = b_0$.

Construct a Rule from the Desired Posteriors

Suppose $\sum_s \alpha_s b_s = b_0$. To obtain the desired signal probabilities and posteriors, we want

$$\mathbb{P}(\theta, s) = \alpha_s b_s(\theta).$$

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We must check that these are valid conditional probabilities and that they produce the promised posteriors.

Converse: Verify the Rule and Its Posteriors

The proposed probabilities are nonnegative. For every state,

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Conditioning on that signal gives the desired belief:

$$\mathbb{P}(\theta \mid s) = \frac{b_0(\theta) \sigma(s \mid \theta)}{\alpha_s} = b_s(\theta).$$

Thus every Bayes-plausible finite split is implementable.

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$$v(b) := \sum_{\theta} b(\theta) u_S(a^*(b), \theta).$$

The state is still averaged according to b ; the receiver's action has been selected. Finite utilities make v bounded.

Choose a Distribution of Posterior Beliefs

Under a signaling scheme, the sender's expected utility is

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$$\sup_{\text{finite schemes } \sigma} \mathbb{E}[u_S] = \sup_{\substack{\text{finite splits } \{(\alpha_s, b_s)\} \\ \sum_s \alpha_s b_s = b_0}} \sum_s \alpha_s v(b_s).$$

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We choose which posterior beliefs occur and how often, subject to their weighted average equaling the prior. What is the best expected payoff achievable under this mean constraint?

A Posterior Determines the Sender's Payoff

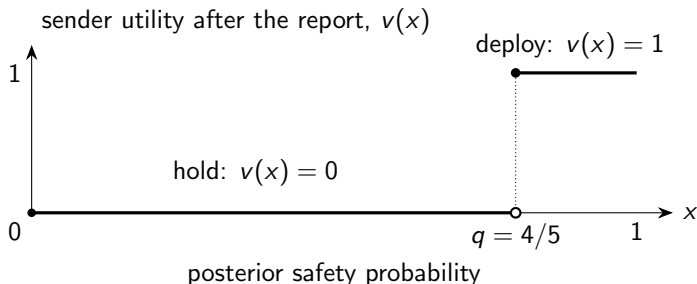
Return to our safe/harmful example. A posterior is summarized by $x = \mathbb{P}(\text{safe} \mid s)$. The user deploys if $x \geq q = 4/5$. The sender earns one for deployment and zero for holding:

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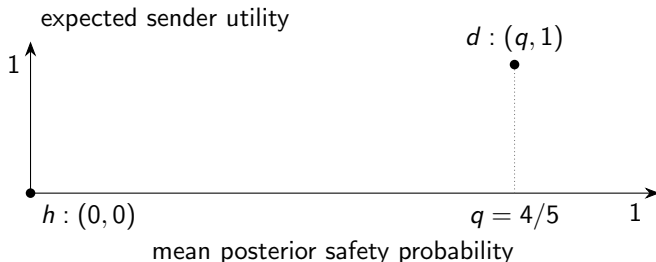
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Each point $(x, v(x))$ describes one possible posterior and the sender's resulting payoff.

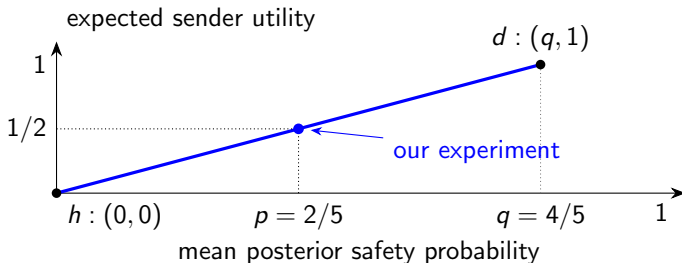
Two Reports Average Two Points on the Graph

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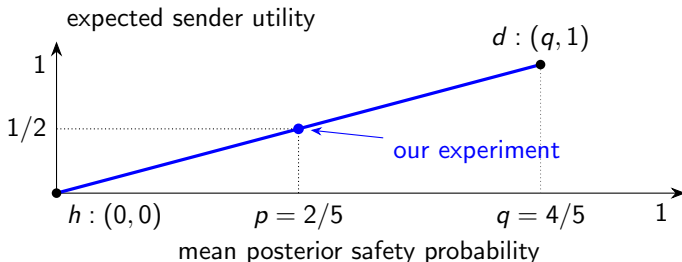
The midpoint records (mean posterior, expected sender utility):

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The mean $p = 2/5$ is not a third possible posterior.

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Varying the probabilities traces the chord. On this chord, our fixed prior permits only the point with mean posterior $p = 2/5$.

A Concave Graph Lies Above Its Chords

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Return to general belief vectors $b, c \in \Delta(\Theta)$. A function g is *concave* if, for $0 \leq t \leq 1$,

$$g(tb + (1 - t)c) \geq tg(b) + (1 - t)g(c).$$

Its graph lies above each chord joining two points on its graph: the value at the mean is at least the average of the values.

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A *concave upper bound* on v is a concave g satisfying $g(b) \geq v(b)$ at every belief. Such a function bounds the payoff at each posterior, and concavity will bound their average payoff. The smallest concave upper bound is the *concave envelope* of v .

The Best Disclosure Value Is the Concave Envelope

Fix a prior b . Define the best expected sender payoff over splits with mean b :

$$V(b) := \sup_{\substack{\text{finite splits } \{(\alpha_s, b_s)\} \\ \sum_s \alpha_s b_s = b}} \sum_s \alpha_s v(b_s).$$

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The splitting lemma identifies $V(b_0)$ with the sender's best expected utility at prior b_0 . We will first show that every concave upper bound g on v satisfies $g(b) \geq V(b)$ for every belief b . We will then show that the function V is itself concave.

Concavity Bounds Every Experiment's Payoff

Let g be concave with $g \geq v$. Take any posterior split $\{(\alpha_s, b_s)\}$ with mean $\sum_s \alpha_s b_s = b$.

Concavity for a finite mixture, called *Jensen's inequality*, gives

$$\begin{aligned} g(b) &= g\left(\sum_s \alpha_s b_s\right) && \text{the split has mean } b \\ &\geq \sum_s \alpha_s g(b_s) && \text{by concavity} \\ &\geq \sum_s \alpha_s v(b_s) && \text{since } g \geq v. \end{aligned}$$

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The last expression is the experiment's expected sender utility. Since the bound holds for every split, taking the supremum gives

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It remains to show that the value function V is itself concave. Then it will be the smallest concave upper bound on v .

Combining Splits Gives a Split of the Average Prior

For concavity, compare priors b, c with their weighted average $tb + (1 - t)c$, where $0 < t < 1$. Take splits of b and c , with expected sender payoffs J_b, J_c :

$$b = \sum_s \alpha_s b_s, \quad J_b = \sum_s \alpha_s v(b_s),$$
$$c = \sum_r \beta_r c_r, \quad J_c = \sum_r \beta_r v(c_r).$$

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Give the first collection total weight t by replacing each α_s with $t\alpha_s$. Give the second total weight $1 - t$ by replacing each β_r with $(1 - t)\beta_r$. The new weights sum to $t + (1 - t) = 1$.

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Give the first collection total weight t by replacing each α_s with $t\alpha_s$. Give the second total weight $1 - t$ by replacing each β_r with $(1 - t)\beta_r$. The new weights sum to $t + (1 - t) = 1$. The combined mean and expected payoff are therefore

$$\text{mean} = \sum_s t\alpha_s b_s + \sum_r (1 - t)\beta_r c_r = tb + (1 - t)c,$$
$$\text{payoff} = \sum_s t\alpha_s v(b_s) + \sum_r (1 - t)\beta_r v(c_r) = tJ_b + (1 - t)J_c.$$

The Best Disclosure Value Is Concave

Fix $0 < t < 1$. For any splits of b and c , with payoffs J_b, J_c , the preceding construction gives a split with mean $tb + (1 - t)c$.

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$$\begin{aligned} V(tb + (1 - t)c) &\geq \sup_{\substack{\text{split of } b \\ \text{split of } c}} [tJ_b + (1 - t)J_c] \\ &= tV(b) + (1 - t)V(c). \end{aligned}$$

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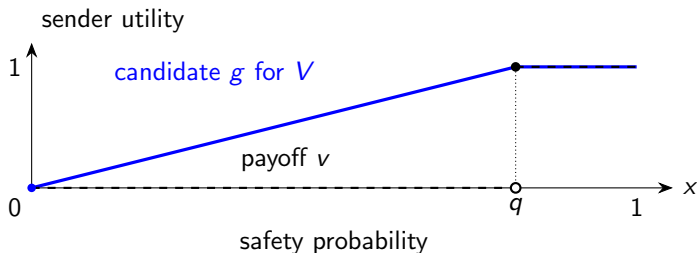
We already showed $V \geq v$ and that every concave upper bound on v lies above V . Thus V is the concave envelope. $\square$

This envelope is also called the *concavification* of v , conventionally written $V = \text{cav } v$.

Finding the Optimal Disclosure Value in Our Example

At posterior safety probability x , the user deploys when $x \geq q$, where $0 < q < 1$. The black graph gives the sender's payoff: $v(x) = 0$ below q and $v(x) = 1$ at or above q .

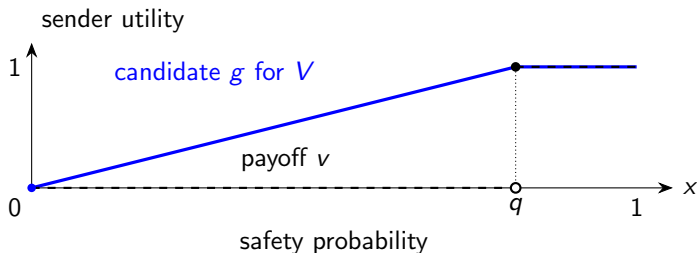
In contrast, $V(p)$ is the best expected payoff starting from prior p : our theorem gives it as the height of the concave envelope at p . The blue curve g is a candidate: it rises linearly from $(0,0)$ to $(q,1)$, then remains at height 1.



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The function g lies above v and is concave: its slope drops from $1/q$ to 0.

The Candidate Curve Is the Concave Envelope

Recall: $g(x) = x/q$ for $x \leq q$, and $g(x) = 1$ for $x \geq q$. We know g is a concave upper bound on v . We must show that every other concave upper bound f lies above g . Since $f \geq v$, we have $f(0) \geq 0$ and $f(q) \geq 1$.

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$$x = \left(1 - \frac{x}{q}\right) 0 + \frac{x}{q} q.$$

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For $x \geq q$, the payoff itself requires $f(x) \geq v(x) = 1 = g(x)$. Thus $f \geq g$ everywhere, so

$$V(x) = g(x) = \min\{x/q, 1\}.$$

Two Reports Achieve the Optimal Deployment Probability

In the binary safe/harmful model, let $0 < p < q < 1$, where p is the prior safety probability. The user deploys exactly when her posterior safety probability is at least q , including equality.

Theorem (Optimal threshold persuasion)

Among all finite signaling rules the sender can commit to, the maximum deployment probability is p/q . A two-report rule attains this maximum.

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Upper bound. Let B be the random posterior safety probability under any such rule. Bayes plausibility gives $\mathbb{E}[B] = p$. On deployment, $B \geq q$; otherwise, $B \geq 0$. Hence

$$p = \mathbb{E}[B] \geq q \mathbb{P}(B \geq q) = q \mathbb{P}(\text{deploy}).$$

No signaling rule can induce deployment more often than p/q .

Two Reports Attain the Bound

Construction. To attain deployment probability p/q , use two reports: one makes the user certain deployment is harmful (posterior 0); the other gives exactly enough confidence to deploy (posterior q).

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$$(1 - \lambda)0 + \lambda q = p \implies \lambda = \frac{p}{q}.$$

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For $p = 2/5$ and $q = 4/5$, this is $1/2$. Our opening disclosure rule is therefore optimal.

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Summary

- ▶ A provider can gain from withholding useful information, even when the user understands its reports and acts optimally.
- ▶ Bayesian updating limits this influence: the provider can change the distribution of posterior beliefs, but not their average. Concavification tells us how much it can gain.
- ▶ Calibration answers whether reports have the right statistical meaning. Alignment also asks whether the choice of information serves the user's interests.

Thanks!

See you next class!

References

- ▶ Emir Kamenica and Matthew Gentzkow. *Bayesian Persuasion*. American Economic Review, 101(6):2590–2615, 2011.